Lctr 12: From Fourier Series to Fourier Transforms (CT and DT)

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Lctr 12: From Fourier Series to Fourier Transforms (Continuous-Time and Discrete-Time)

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Focus: Sections 4.0–4.2 and 5.0–5.2 (Pages 284–296, 358–374)

12.1 The Big Picture: Beyond Periodic Signals

Why This Matters

In Lectures 9–11 we developed the **Fourier series**, which decomposes *periodic* signals into sums of harmonically related complex exponentials. But most real-world signals are *not* periodic: a spoken word, a radar pulse, or a transient voltage spike all have finite duration. How can we analyze these signals in the frequency domain? The answer is the **Fourier transform**—a generalization of the Fourier series that works for *any* signal (periodic or aperiodic), as long as it has finite energy or satisfies certain convergence conditions. This lecture derives the Fourier transform for both continuous-time (CT) and discrete-time (DT) signals, and establishes the basic transform pairs that we will use throughout the rest of the course.

Roadmap for this lecture:

1. Derive the CT Fourier Transform by taking the Fourier series to the limit $T \to \infty$.
2. Establish the CT Fourier Transform pair and compute basic transform pairs.
3. Derive the DT Fourier Transform (DTFT) in parallel, by taking $N \to \infty$.
4. Establish the DTFT pair and compute basic DT transform pairs.
5. Show how the Fourier Transform handles periodic signals using impulse trains.
6. Summarize everything in a side-by-side CT vs. DT comparison table.

12.2 From CT Fourier Series to CT Fourier Transform

12.2.1 The Key Idea: Let the Period Go to Infinity

Suppose we have an aperiodic signal $x(t)$ that is nonzero only over some finite interval (for example, a rectangular pulse). We can "embed" this signal inside a periodic signal $\tilde{x}(t)$ with period T by simply repeating $x(t)$ every T seconds.

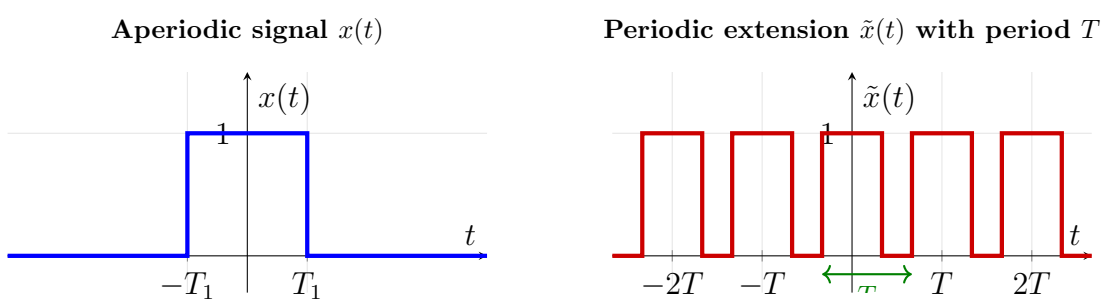

Figure 1: Left: An aperiodic rectangular pulse $x(t)$. Right: Its periodic extension $\tilde{x}(t)$ with period T. As $T \to \infty$, the copies move infinitely far apart and $\tilde{x}(t) \to x(t)$.

The periodic signal $\tilde{x}(t)$ *does* have a Fourier series. Its coefficients are:

$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} \tilde{x}(t)\, e^{-jk\omega_0 t}\, dt = \frac{1}{T}\int_{-T/2}^{T/2} x(t)\, e^{-jk\omega_0 t}\, dt$$

where $\omega_0 = 2\pi/T$ is the fundamental frequency.

Now define the **envelope function**:

$$X(j\omega) \stackrel{\text{def}}{=} \int_{-\infty}^{+\infty} x(t)\, e^{-j\omega t}\, dt$$

Since $x(t)$ is nonzero only on a finite interval (or decays fast enough), the limits $\pm T/2$ can be replaced by $\pm\infty$ for large enough T. This gives us:

$$\boxed{a_k = \frac{1}{T}\, X(jk\omega_0)}$$

Key Insight

The Fourier series coefficients a_k are **samples** of the envelope $X(j\omega)$, evaluated at the harmonic frequencies $\omega = k\omega_0$, and scaled by $1/T$. As T grows, the harmonics $k\omega_0$ get closer together (because $\omega_0 = 2\pi/T$ shrinks), and the samples become denser.

12.2.2 Visualizing the Limiting Process

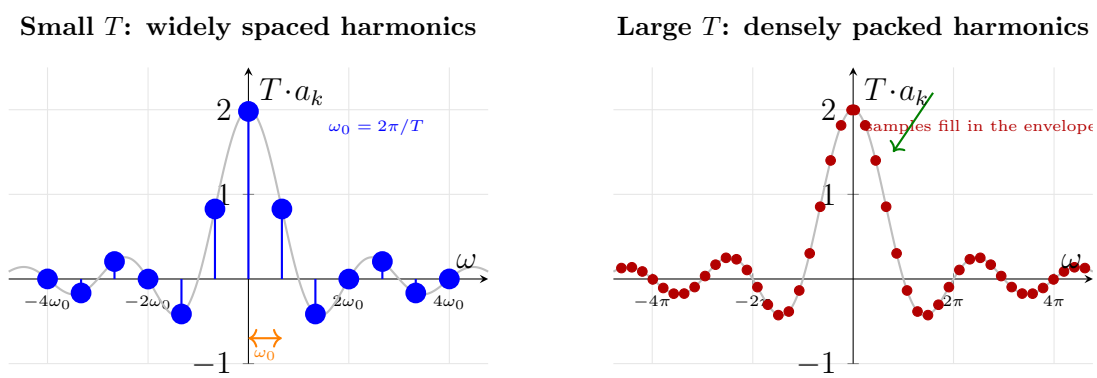

Figure 2: As T increases, the spacing $\omega_0 = 2\pi/T$ between harmonics shrinks. The scaled coefficients $T \cdot a_k = X(jk\omega_0)$ become denser samples of the *same* envelope $X(j\omega)$. In the limit $T \to \infty$, the samples fill in the entire continuous curve.

12.2.3 Taking the Limit: The Sum Becomes an Integral

The Fourier series synthesis equation is:

$$\tilde{x}(t) = \sum_{k=-\infty}^{+\infty} a_k\, e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} \frac{1}{T}\, X(jk\omega_0)\, e^{jk\omega_0 t} = \frac{1}{2\pi}\sum_{k=-\infty}^{+\infty} X(jk\omega_0)\, e^{jk\omega_0 t}\, \omega_0$$

where we used $1/T = \omega_0/(2\pi)$ in the last step. As $T \to \infty$:

- $\omega_0 \to d\omega$ (the spacing becomes infinitesimal),
- $k\omega_0 \to \omega$ (a continuous frequency variable),
- the sum $\to$ an integral.

This gives us the **CT Fourier Transform synthesis equation**:

$$\boxed{x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega)\, e^{j\omega t}\, d\omega \qquad \textbf{(Inverse Fourier Transform)}}$$

The **analysis equation** is the envelope function we defined earlier:

$$\boxed{X(j\omega) = \int_{-\infty}^{+\infty} x(t)\, e^{-j\omega t}\, dt \qquad \textbf{(Fourier Transform)}}$$

Key Insight

The Fourier Transform pair generalizes the Fourier series:

- **Analysis** (forward transform): decomposes a signal into a **continuous** spectrum $X(j\omega)$.
- **Synthesis** (inverse transform): reconstructs the signal from its spectrum.

The notation $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$ means "$x(t)$ and $X(j\omega)$ are a Fourier transform pair."

12.2.4 Convergence of the CT Fourier Transform

Just like the Fourier series, the Fourier transform requires convergence conditions. The Dirichlet conditions for the FT are:

1. $x(t)$ is **absolutely integrable**: $\int_{-\infty}^{+\infty} |x(t)|\, dt < \infty$
2. $x(t)$ has a finite number of maxima, minima, and discontinuities in every finite interval.

Alternatively, a sufficient condition is that $x(t)$ has **finite energy**: $\int_{-\infty}^{+\infty} |x(t)|^2\, dt < \infty$.

Common Mistake

Periodic signals like $\cos(\omega_0 t)$ are **not** absolutely integrable and do **not** have finite energy. Their Fourier transforms involve **impulse functions** $\delta(\omega)$, which we will address in Section 12.4.

12.3 Basic CT Fourier Transform Pairs

Let us compute the Fourier transforms of several important signals from scratch.

12.3.1 One-Sided Decaying Exponential: $x(t) = e^{-at}u(t), \quad a > 0$

Fourier Transform of the One-Sided Decaying Exponential

Step 1: Apply the definition. Since $u(t) = 0$ for $t < 0$:

$$X(j\omega) = \int_{-\infty}^{+\infty} e^{-at}u(t)\, e^{-j\omega t}\, dt = \int_0^{\infty} e^{-at}\, e^{-j\omega t}\, dt = \int_0^{\infty} e^{-(a+j\omega)t}\, dt$$

Step 2: Evaluate the integral. Since $a > 0$, the exponential decays:

$$X(j\omega) = \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)}\right]_0^{\infty} = 0 - \frac{1}{-(a+j\omega)} = \frac{1}{a+j\omega}$$

Step 3: Extract magnitude and phase:

$$|X(j\omega)| = \frac{1}{\sqrt{a^2+\omega^2}}, \qquad \angle X(j\omega) = -\arctan\left(\frac{\omega}{a}\right)$$

Numerical check ($a = 2$):

- At $\omega = 0$: $|X| = 1/2 = 0.5$
- At $\omega = 2$ $(= a)$: $|X| = 1/\sqrt{8} \approx 0.354, \quad \angle X = -45°$
- At $\omega = 10$: $|X| = 1/\sqrt{104} \approx 0.098$

The spectrum decays toward zero as $|\omega| \to \infty$—most of the energy is concentrated at low frequencies.

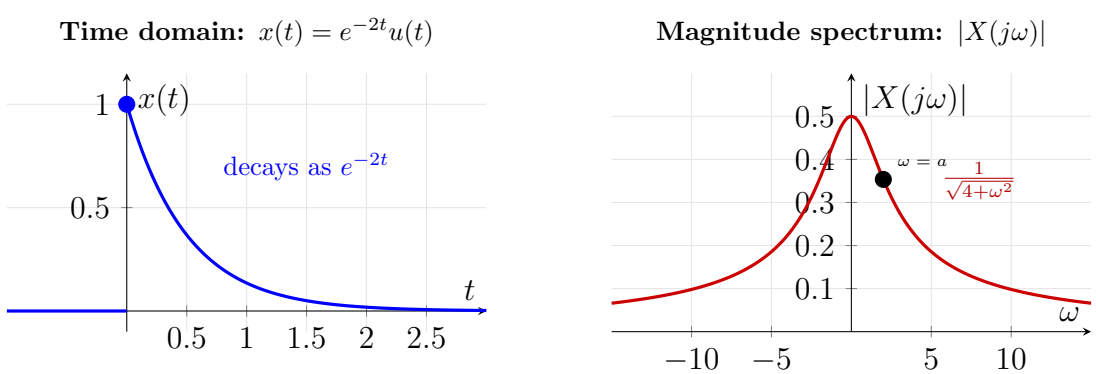

Figure 3: Fourier transform pair: $e^{-2t}u(t) \overset{\mathcal{F}}{\longleftrightarrow} 1/(2+j\omega)$. The faster the signal decays in time (larger a), the *wider* the spectrum—another manifestation of the time–frequency trade-off.

12.3.2 Two-Sided Decaying Exponential: $x(t) = e^{-a|t|}, \quad a > 0$

Fourier Transform of the Two-Sided Decaying Exponential

Step 1: Split the integral at $t = 0$:

$$X(j\omega) = \int_{-\infty}^{0} e^{at}\, e^{-j\omega t}\, dt + \int_{0}^{\infty} e^{-at}\, e^{-j\omega t}\, dt$$

Step 2: Evaluate each half:

$$\int_{-\infty}^{0} e^{(a-j\omega)t}\, dt = \frac{1}{a - j\omega}$$
$$\int_{0}^{\infty} e^{-(a+j\omega)t}\, dt = \frac{1}{a + j\omega}$$

Step 3: Add them:

$$X(j\omega) = \frac{1}{a - j\omega} + \frac{1}{a + j\omega} = \frac{(a + j\omega) + (a - j\omega)}{a^2 + \omega^2} = \boxed{\frac{2a}{a^2 + \omega^2}}$$

This is **purely real** and positive for all ω (because $x(t)$ is real and even).

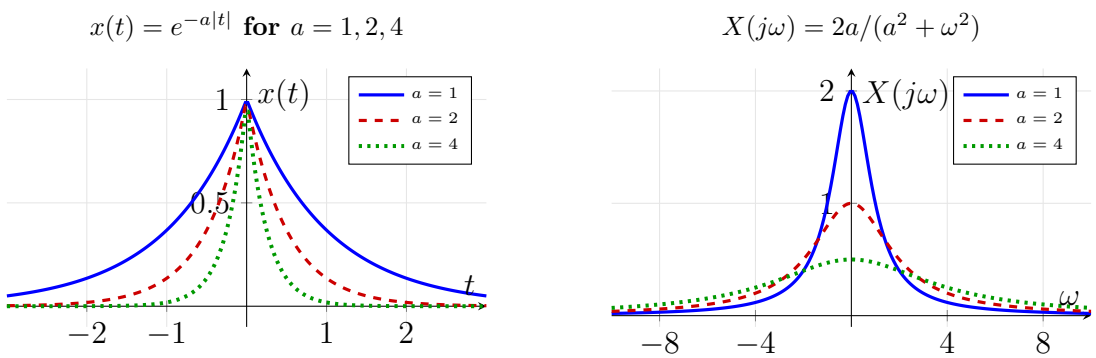

Figure 4: The time–frequency trade-off in action. Larger $a \Rightarrow$ faster decay in time (narrower signal) but *wider* spectrum. Smaller $a \Rightarrow$ slower decay (wider signal) but *narrower* spectrum.

12.3.3 The Unit Impulse: $x(t) = \delta(t)$

Fourier Transform of the Unit Impulse

$$X(j\omega) = \int_{-\infty}^{+\infty} \delta(t)\, e^{-j\omega t}\, dt = e^{-j\omega \cdot 0} = 1$$

Result: $\delta(t) \overset{\mathcal{F}}{\longleftrightarrow} 1$
The impulse contains **all frequencies equally**. Its spectrum is flat and constant at 1 across all of frequency.

Key Insight

Think of it this way: the impulse is infinitely sharp in time, so by the time–frequency trade-off, it must be infinitely wide in frequency. A "flash" in time excites every frequency simultaneously.

12.3.4 The Rectangular Pulse: $x(t) = \begin{cases} 1, & |t| \le T_1 \\ 0, & |t| > T_1 \end{cases}$

Fourier Transform of a Rectangular Pulse

Step 1: Apply the definition:

$$X(j\omega) = \int_{-T_1}^{T_1} 1 \cdot e^{-j\omega t}\, dt = \left[\frac{e^{-j\omega t}}{-j\omega}\right]_{-T_1}^{T_1} = \frac{e^{-j\omega T_1} - e^{j\omega T_1}}{-j\omega}$$

Step 2: Recognize the sine (using $e^{j\theta} - e^{-j\theta} = 2j\sin\theta$):

$$X(j\omega) = \frac{-2j\sin(\omega T_1)}{-j\omega} = \frac{2\sin(\omega T_1)}{\omega}$$

Step 3: At $\omega = 0$, use L'Hôpital's rule (or just note that the integral gives $2T_1$):

$$X(j \cdot 0) = 2T_1$$

Numerical example ($T_1 = 1$):

- $X(j \cdot 0) = 2$ (peak value)
- $X(j\pi) = 2\sin(\pi)/\pi = 0$ (first zero)
- $X(j \cdot 3\pi/2) = 2\sin(3\pi/2)/(3\pi/2) = -4/(3\pi) \approx -0.424$

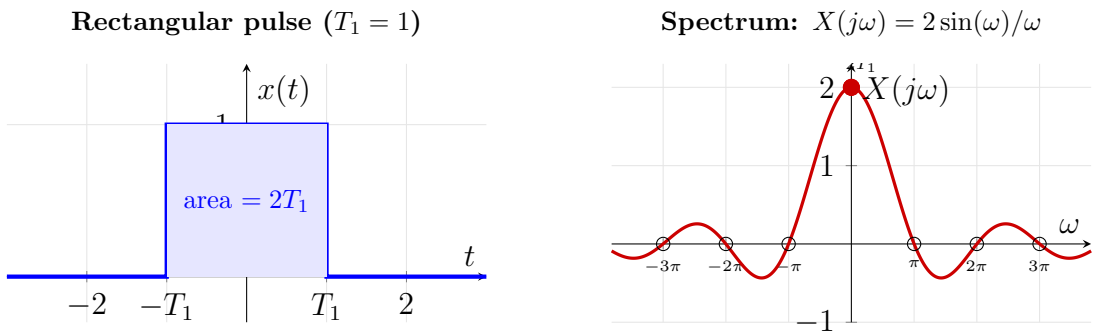

Figure 5: A rectangular pulse in time has a $\sin(\omega T_1)/\omega$ spectrum (a "sinc-like" shape). The zeros occur at $\omega = n\pi/T_1$ for integer $n \neq 0$. Wider pulse $\Rightarrow$ narrower main lobe (time–frequency trade-off again).

Common Mistake

There are two common conventions for the "sinc" function:

- **Unnormalized:** $\text{sinc}(x) = \sin(x)/x$ (used by Oppenheim)
- **Normalized:** $\text{sinc}(x) = \sin(\pi x)/(\pi x)$ (used in MATLAB and some other textbooks)

In this course, we follow Oppenheim's convention. Always check which convention is being used when consulting different sources or software.

12.4 CT Fourier Transform of Periodic Signals

Why This Matters

We said the Fourier transform works for aperiodic signals. But what about periodic signals—can we still use the FT framework? Yes! The trick is to use impulse functions $\delta(\omega)$ in the frequency domain. This unifies periodic and aperiodic signals under a single framework.

12.4.1 The FT of a Complex Exponential

Consider $x(t) = e^{j\omega_0 t}$. This signal is *not* absolutely integrable, so we cannot compute its FT by direct integration. Instead, we work backwards from the inverse FT. If the spectrum were $X(j\omega) = 2\pi\,\delta(\omega - \omega_0)$, then:

$$\frac{1}{2\pi}\int_{-\infty}^{+\infty} 2\pi\,\delta(\omega - \omega_0)\,e^{j\omega t}\,d\omega = e^{j\omega_0 t} \quad \checkmark$$

Therefore:

$$\boxed{e^{j\omega_0 t} \overset{\mathcal{F}}{\longleftrightarrow} 2\pi\,\delta(\omega - \omega_0)}$$

12.4.2 The FT of Cosine and Sine

Using Euler's formula:

$$\cos(\omega_0 t) = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \overset{\mathcal{F}}{\longleftrightarrow} \pi\big[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)\big]$$

$$\sin(\omega_0 t) = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} \overset{\mathcal{F}}{\longleftrightarrow} \frac{\pi}{j}\big[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)\big]$$

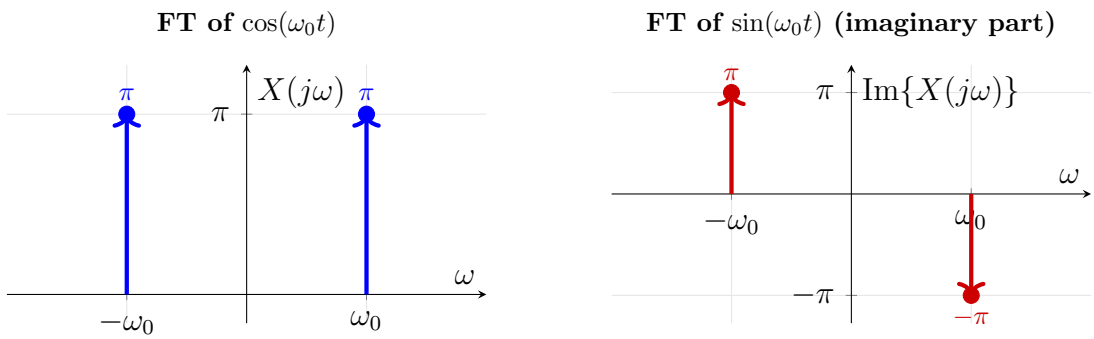

Figure 6: The FT of $\cos(\omega_0 t)$ consists of two equal impulses at $\pm\omega_0$ (real-valued). The FT of $\sin(\omega_0 t)$ consists of two impulses of opposite sign along the imaginary axis.

12.4.3 General Periodic Signals

For *any* periodic signal with Fourier series $x(t) = \sum_k a_k\, e^{jk\omega_0 t}$, linearity gives:

$$\boxed{X(j\omega) = \sum_{k=-\infty}^{+\infty} 2\pi\, a_k\, \delta(\omega - k\omega_0)}$$

Key Insight

The Fourier transform of a periodic signal is a **train of impulses** in the frequency domain. The impulse at $\omega = k\omega_0$ has area ("weight") $2\pi a_k$. This is the bridge between Fourier series and the Fourier transform: the FS coefficients become the weights of the impulses.

12.5 The Discrete-Time Fourier Transform (DTFT)

Why This Matters

The development of the DT Fourier transform follows **exactly the same logic** as the CT case: start with the DT Fourier series, let the period $N \to \infty$, and watch the sum become an integral. If you understood Sections 12.2–12.4, this section will feel like a parallel universe with the same structure but slightly different notation.

12.5.1 From DT Fourier Series to DTFT

For a DT periodic signal $\tilde{x}[n]$ with period N, the DT Fourier series coefficients are:

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}[n]\, e^{-jk(2\pi/N)n}$$

Define the **envelope**:

$$X(e^{j\omega}) \stackrel{\text{def}}{=} \sum_{n=-\infty}^{+\infty} x[n]\, e^{-j\omega n}$$

Then $a_k = \frac{1}{N} X(e^{j2\pi k/N})$. As $N \to \infty$, the DT FS synthesis sum becomes an integral over ω, giving us the **DTFT pair**:

$$\boxed{X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]\, e^{-j\omega n} \qquad \textbf{(DTFT — Analysis)}}$$

$$\boxed{x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega \qquad \textbf{(Inverse DTFT — Synthesis)}}$$

where $\int_{2\pi}$ denotes integration over any interval of length 2π (typically $-\pi$ to π or 0 to 2π).

12.5.2 Key Difference from CT: Periodicity in Frequency

Key Insight

The DTFT $X(e^{j\omega})$ is periodic in ω with period 2π.
This is a fundamental difference from the CT Fourier Transform (which is *not* periodic in ω). The reason: $e^{j(\omega+2\pi)n} = e^{j\omega n} \cdot e^{j2\pi n} = e^{j\omega n}$ for integer n.
Consequence: we only need to plot $X(e^{j\omega})$ over one period, typically $-\pi \le \omega \le \pi$.
Low DT frequencies are near $\omega = 0$; high DT frequencies are near $\omega = \pm\pi$.

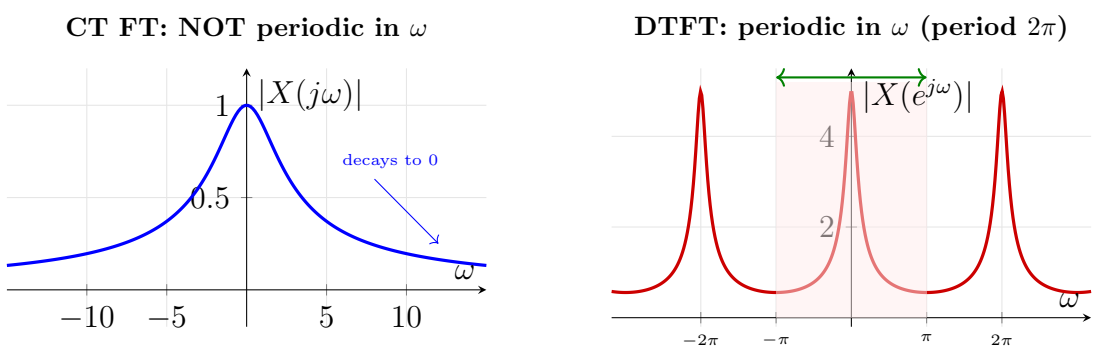

Figure 7: Left: A typical CT Fourier transform decays to zero as $|\omega| \to \infty$. Right: The DTFT repeats every 2π—the shaded region is one period.

12.5.3 Convergence of the DTFT

The DTFT converges if $x[n]$ is **absolutely summable**: $\sum_{n=-\infty}^{+\infty} |x[n]| < \infty$, or if $x[n]$ has **finite energy**: $\sum_{n=-\infty}^{+\infty} |x[n]|^2 < \infty$. No Gibbs-type phenomenon arises in DTFT synthesis because the inverse DTFT is a continuous integral (not a truncated sum).

12.6 Basic DTFT Pairs

12.6.1 The Unit Impulse: $x[n] = \delta[n]$

DTFT of the Unit Impulse

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} \delta[n]\, e^{-j\omega n} = e^{-j\omega \cdot 0} = 1$$

Result: $\delta[n] \overset{\mathcal{F}}{\longleftrightarrow} 1$
Exactly the same as in CT! The discrete-time impulse also contains all frequencies equally.

12.6.2 DT Decaying Exponential: $x[n] = a^n u[n], \quad |a| < 1$

DTFT of the DT Decaying Exponential

Step 1: Apply the definition:

$$X(e^{j\omega}) = \sum_{n=0}^{\infty} a^n \, e^{-j\omega n} = \sum_{n=0}^{\infty} (ae^{-j\omega})^n$$

Step 2: This is a geometric series with ratio $r = ae^{-j\omega}$. Since $|r| = |a| < 1$:

$$X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$$

Step 3: Magnitude:

$$|X(e^{j\omega})| = \frac{1}{|1 - ae^{-j\omega}|} = \frac{1}{\sqrt{(1 - a\cos\omega)^2 + (a\sin\omega)^2}} = \frac{1}{\sqrt{1 + a^2 - 2a\cos\omega}}$$

Numerical check ($a = 0.8$):

- At $\omega = 0$: $|X| = 1/(1 - 0.8) = 5$ (maximum)
- At $\omega = \pi/2$: $|X| = 1/\sqrt{1 + 0.64} = 1/\sqrt{1.64} \approx 0.781$
- At $\omega = \pi$: $|X| = 1/(1 + 0.8) = 1/1.8 \approx 0.556$ (minimum)

The spectrum peaks at $\omega = 0$: this is a **lowpass** spectrum (same signal, by the way, as the impulse response of the first-order recursive filter from Lecture 11!).

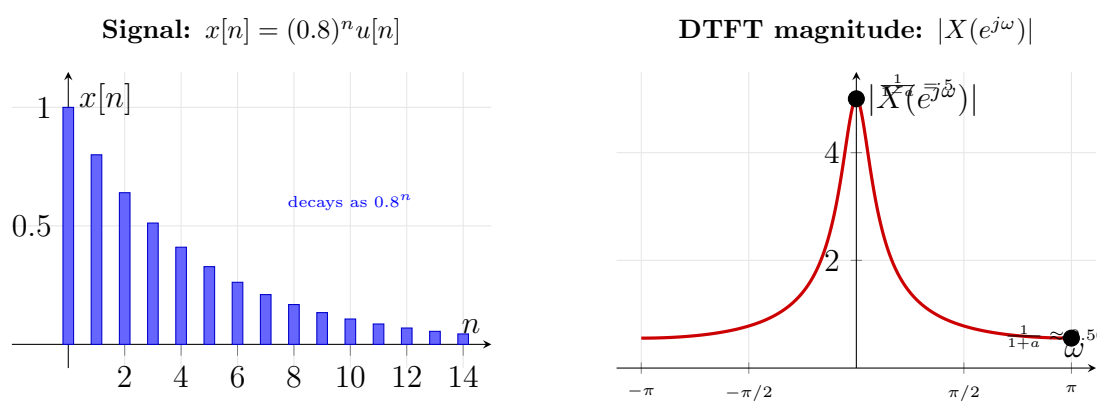

Figure 8: DTFT pair: $0.8^n u[n] \overset{\mathcal{F}}{\longleftrightarrow} 1/(1 - 0.8\, e^{-j\omega})$. The spectrum is lowpass-shaped, peaking at $\omega = 0$. Only one period ($-\pi$ to π) is shown.

12.6.3 Finite-Length Sequence: $x[n] = \begin{cases} 1, & 0 \le n \le M \\ 0, & \textbf{otherwise} \end{cases}$

DTFT of a Rectangular Window

Step 1: Apply the definition:

$$X(e^{j\omega}) = \sum_{n=0}^{M} e^{-j\omega n} = \frac{1 - e^{-j\omega(M+1)}}{1 - e^{-j\omega}}$$

(finite geometric series with $M + 1$ terms).
Step 2: Factor to get a nicer form:

$$X(e^{j\omega}) = e^{-j\omega M/2} \frac{\sin\big(\omega(M+1)/2\big)}{\sin(\omega/2)}$$

Numerical example ($M = 4$, so 5 samples): $X(e^{j0}) = 5$ (sum of five 1's). The first zero occurs when $\omega(M + 1)/2 = \pi$, i.e., $\omega = 2\pi/5$.

This is the DT analog of the rectangular pulse sinc: a $\sin(N\omega/2)/\sin(\omega/2)$ pattern, sometimes called a **Dirichlet kernel**. It is periodic in ω with period 2π, as expected.

DTFT magnitude of a length-5 rectangular window ($M = 4$)

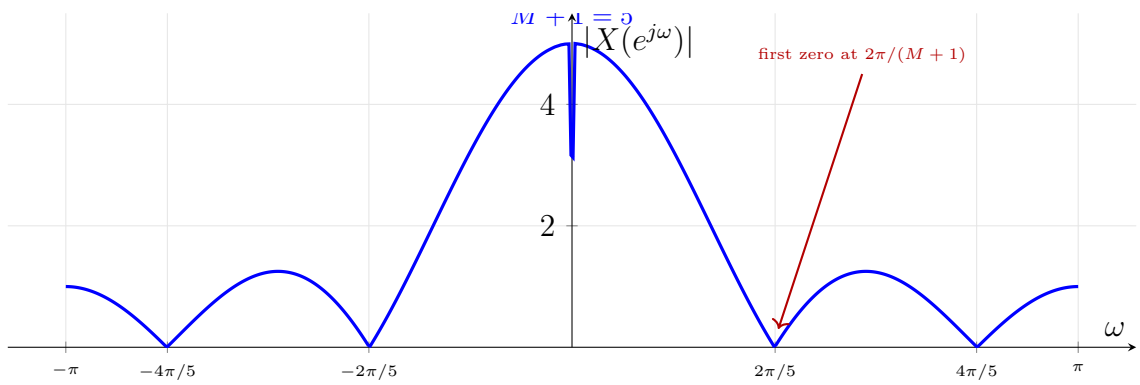

Figure 9: The DTFT of a rectangular window of length 5. The main lobe has width $4\pi/(M+1) = 4\pi/5$, and there are sidelobes that decay slowly. Longer windows $\Rightarrow$ narrower main lobe (better frequency resolution).

12.7 DTFT of Periodic Signals

Just as in the CT case, periodic DT signals have DTFTs consisting of **impulse trains** in the frequency domain. If $x[n]$ is periodic with period N and DT Fourier series coefficients

a_k, then:

$$\boxed{X(e^{j\omega}) = \sum_{k=\langle N\rangle} 2\pi\, a_k\, \delta\left(\omega - \frac{2\pi k}{N}\right), \quad \text{repeated with period } 2\pi}$$

DTFT of a DT Cosine

Let $x[n] = \cos(2\pi n/N)$. Its DT FS coefficients are $a_1 = a_{-1} = 1/2$, with all others zero. Therefore:

$$X(e^{j\omega}) = \pi\,\delta\left(\omega - \frac{2\pi}{N}\right) + \pi\,\delta\left(\omega + \frac{2\pi}{N}\right) + \text{periodic repetitions every } 2\pi$$

This is exactly analogous to the CT result: a cosine has two impulses at $\pm\omega_0$ in its FT.

Key Insight

The connection between DT FS coefficients and the DTFT is the same as in CT: the FS coefficients a_k become the weights of the impulses in the transform. This unifies the periodic and aperiodic worlds under one framework.

12.8 Summary: CT Fourier Transform vs. DTFT

	Continuous-Time FT	**Discrete-Time FT (DTFT)**
Analysis	$X(j\omega) = \int_{-\infty}^{+\infty} x(t)\, e^{-j\omega t}\, dt$	$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]\, e^{-j\omega n}$
Synthesis	$x(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} X(j\omega)\, e^{j\omega t}\, d\omega$	$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$
Domain of ω	$-\infty < \omega < +\infty$	Periodic, period 2π
Convergence	$\int \lvert x(t)\rvert\, dt < \infty$	$\sum \lvert x[n]\rvert < \infty$
Periodic?	$X(j\omega)$: NO	$X(e^{j\omega})$: YES (2π)
Impulse	$\delta(t) \leftrightarrow 1$	$\delta[n] \leftrightarrow 1$
Decay. exp.	$e^{-at}u(t) \leftrightarrow \frac{1}{a+j\omega}$	$a^n u[n] \leftrightarrow \frac{1}{1-ae^{-j\omega}}$
Rect. pulse	$\leftrightarrow \frac{2\sin(\omega T_1)}{\omega}$	$\leftrightarrow e^{-j\omega M/2}\frac{\sin\left(\frac{(M+1)\omega}{2}\right)}{\sin(\omega/2)}$
Periodic sig.	$\sum_k 2\pi a_k\, \delta(\omega - k\omega_0)$	$\sum_k 2\pi a_k\, \delta\left(\omega - \frac{2\pi k}{N}\right)$

12.9 The Big Picture: How Everything Connects

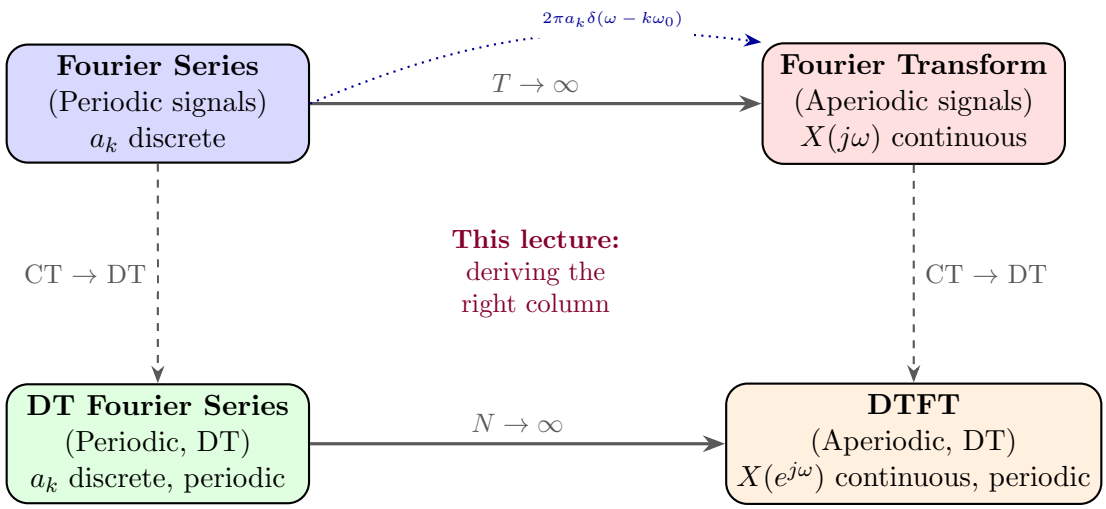

Figure 10: The four Fourier representations and how they relate. This lecture derives the Fourier transforms (right column) from the Fourier series (left column) by letting the period go to infinity. The periodic-signal bridge (dotted arrow) unifies both under the FT framework using impulses.

12.10 Common Mistakes to Avoid

1. **Confusing FS coefficients with the FT:** The FS gives you *discrete* coefficients a_k (one per harmonic). The FT gives you a *continuous function* $X(j\omega)$. They are related by $a_k = \frac{1}{T}X(jk\omega_0)$ for periodic signals, but they are not the same object.

2. **Forgetting the $1/(2\pi)$ in the inverse FT:** The CT synthesis equation has a $1/(2\pi)$ factor that the analysis equation does not. This asymmetry is a convention choice (Oppenheim's convention). Always double-check which equation you are using.

3. **Forgetting that the DTFT is periodic in ω:** Unlike the CT Fourier transform, $X(e^{j\omega})$ repeats every 2π. When plotting, you typically show just $-\pi \leq \omega \leq \pi$. The CT Fourier transform $X(j\omega)$ does *not* repeat.

4. **Wrong sign in the exponent:** The *analysis* (forward) transform uses $e^{-j\omega t}$ (negative sign). The *synthesis* (inverse) uses $e^{+j\omega t}$ (positive sign). Swapping the sign gives you the wrong answer.

5. **Plugging $\omega = 0$ into expressions with ω in the denominator:** For example, $X(j\omega) = 2\sin(\omega T_1)/\omega$. At $\omega = 0$, you must use L'Hôpital or the original integral to get $X(0) = 2T_1$. Do not write "0/0" and give up.

6. **Treating the DTFT geometric series as valid for $|a| \geq 1$:** The sum $\sum_{n=0}^{\infty}(ae^{-j\omega})^n$ converges only when $|a| < 1$. If $|a| \geq 1$, the DTFT does not exist in the ordinary sense (the signal has infinite energy).

Key Insight

Quick sanity checks:

- The value of $X(j\omega)$ at $\omega = 0$ equals $\int x(t)\,dt$ (total area of the signal). Similarly, $X(e^{j0}) = \sum x[n]$ (total sum).
- If $x(t)$ is real and even, then $X(j\omega)$ is real and even.
- If $x(t)$ is real and odd, then $X(j\omega)$ is purely imaginary and odd.

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